NGUYỄN KẾ SANG

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PROFESSIONAL SUMMARY

AI/ML Developer with 6 months of practical experience in training and fine-tuning ML/DL models, complemented by hands-on backend and UI development skills. Highly skilled in data preparation, data processing, and building solutions for OCR, Computer Vision, and NLP.

WORK EXPERIENCE

Mecsu Technical Department

AI/ML Developer (6 Months Experience) *Mar 2026 – Oct 2026*

- Designed and developed prompts and logical workflows for an intelligent AI Chatbot system.
- Handled end-to-end data preparation and processing pipelines for diverse data formats, including images, text, and PDF documents.
- Trained and fine-tuned Machine Learning and Deep Learning models for accurate image classification and text categorization.
- Gained practical full-stack experience by developing robust backend APIs and designing intuitive UI dashboards for business data visualization and AI model integration.

CORE SKILLS

AI & Machine Learning:	Model Training & Fine-tuning, Transfer Learning, Optical Character Recognition (OCR), Computer Vision, Natural Language Processing (NLP), CNN, RNN, LSTM, Transformers, Generative AI.
Data Pipelines & Libs:	PyTorch, Hugging Face, OpenCV, Tesseract OCR, Pandas, NumPy, Scikit-learn, Data Pipeline Construction, Data Augmentation, Data Annotation & Preprocessing.
Tools & Platforms:	Docker, AWS, MLflow, Label Studio / CVAT, GitHub, Google Cloud API, FastAPI, SQL Server, SQLite.
Programming Languages:	Python, C#, Java, SQL, JavaScript, TypeScript.

HIGHLIGHTED PROJECTS

Text-to-SQL Business System

Technologies: RAG, LLM APIs, Prompt Engineering, SQLite, NLP

- **Situation & Task:** Non-technical users needed a way to query business data without knowing SQL.
- **Action:** Developed an AI-powered system converting Vietnamese business questions into executable SQL queries using RAG architecture, schema analysis, and entity recognition.
- **Result:** Achieved high accuracy in SQL validation and query execution, significantly accelerating real-time data retrieval for business decision-making.

Mecsu Image Search & Classification

Technologies: PyTorch, Computer Vision, CNN, Deep Learning

- **Situation & Task:** Required a robust system to predict labels and categorize products based purely on visual inputs.
- **Action:** Applied computer vision techniques and trained custom deep learning models to process large image datasets, extracting core visual features for classification.
- **Result:** Delivered a highly accurate image search system allowing users to upload queries and retrieve precise predicted labels based on visual content.

Visual Question Answering (VQA) System

Technologies: Transformers, CNN, Deep Learning, NLP

- **Task:** Build a system capable of understanding an image and answering natural language questions about it.
- **Action:** Integrated CNNs for object detection/feature extraction and Transformers for NLP context understanding to analyze visual content.
- **Result:** Successfully generated contextually appropriate responses based on detected objects, bridging the gap between Computer Vision and NLP.

Deep Learning Research: NLP & Computer Vision

Technologies: Seq2Seq, Transformers, RNN, Class Activation Map (CAM)

- **Action (NLP):** Researched and implemented machine translation and chatbot Q&A models using Seq2Seq, RNN, and Transformers.
- **Action (CV):** Generated counterfactual images using CAM to deeply analyze, interpret, and explain deep learning model predictions.
- **Result:** Gained comprehensive expertise in model explainability and fine-tuning state-of-the-art architectures for both text and image tasks.

EDUCATION & CERTIFICATIONS

Ton Duc Thang University

Degree: Bachelor of Computer Science (Graduated 2026) — GPA: 7.3/10

Aptis ESOL – British Council (2022)

English language proficiency certificate assessing listening, speaking, reading, and writing.